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Lightweight visual backbone network with enhanced comprehensive strength through context-aware dual attention mechanism

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ABSTRACT

Recent studies have increasingly concentrated on integrating visual neural networks into edge devices, especially for robots and drones that require effective real-time decision-making and autonomous path planning. Achieving this necessitates lightweight visual networks with robust overall performance. In this study, we adopt a small-scale architecture based on convolutional neural networks (CNNs) and propose a Context-Aware Decoupled Fully Connected (CADFC) attention mechanism to enhance the performance of existing CNN-based visual networks. The core concept of the CADFC attention mechanism is to incorporate both remote and local contexts, enabling the network to capture local features and their relevant remote features. Moreover, we design a novel bottleneck that integrates CADFC attention with depthwise convolution. This design thinking is to accumulate focused regions, learning from depthwise convolution, alongside their surrounding features based on CADFC attention. Experimental results demonstrate that CADFC MobileNet outperforms the recent SOTA network GhostNetV2, achieving a top-1 accuracy of 76.8% with 15% fewer parameters, surpassing GhostNetV2's 75.3% accuracy by 1.5%, even in resource-constrained environments.

1. Introduction

Recent studies have increasingly concentrated on integrating visual neural networks into edge devices, particularly for applications like motion simulation in robotics [1,2], robot soccer ball tracking [3,4] and faster obstacle avoidance in drones [5,6]. In these advanced applications, visual neural networks must have strong comprehensive strength to enable robots and drones to effectively make real-time decisions and plan paths. For example, the network must facilitate movement through image classification [7] and object detection [8] at varying distances, allowing for the recognition of the distance between remote objects, such as balls, and the robot, as well as the angles between body segments and local joints for precise maneuvers.

To improve the comprehensive strength of visual neural networks, a common approach is the integration of self-attention mechanisms [9–12] into large-scale architectures like Vision Transformers (ViT) [13] and Masked Autoencoders (MAE) [14]. These self-attention mechanisms enable visual networks to assign varying levels of importance to different spatial regions, allowing them to selectively emphasize relevant features while diminishing the impact of less significant ones, thereby enhancing overall performance.

Due to resource limitations, edge devises often deployed lightweight visual networks, such as MobileNet [15] and ShuffleNet [16], instead of large-scale networks. In these lightweight networks, the classic convolutional neural networks (CNNs), i.e., AlexNet [17], VGGNet [18], GoogleNet [19], ResNet [20], and EfficientNet [21], are used as the foundational model for end-to-end solving various visual tasks, i.e., imagine classification, object detection [22,8]. However, existing CNN-based mobile networks often focus predominantly on local features, relying on convolutional layers that process nearby pixels. This emphasis on local information can limit their ability to address fundamental tasks, resulting in reduced overall performance.

In this study, inspired by the Decoupled Fully Connected (DFC) method, we present a novel context-aware DFC attention mechanism, termed CADFC, designed to capture both local and remote key information within each input feature map, which is segmented into non-overlapping uniform patches. The CADFC mechanism decouples the key information in a full feature map into local and remote contexts. The local context captures dependencies among elements within a patch, while the remote context focuses on dependencies among elements across different patches. The CADFC mechanism constructs a vertical attention map that processes information in the vertical

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direction based on remote context, as well as a horizontal attention map that processes information in the horizontal direction based on local context. This dual attention approach broadens the perspective, allowing for a comprehensive consideration of a wider range of contextual elements.

Taking advantage of CADFC dual attention mechanism, we further propose a novel visual backbone network CADFC MobileNet, to adapt the edge devices with strong comprehensive strength. This backbone network is derived from MobileNetV2 network, and optimized at two aspects: (1) design a CADFC bottleneck block to fusion the feature information from deep convolutional kernels and CADFC attention mechanism to improve feature extraction capabilities of convolution operations, (2) downsample the input features by half and apply bilinear interpolation to upsample the attention map, ensuring it aligns in size with the feature map to reduce computational complexity. Extensive experiments on the ImageNet dataset [7] for image classification, MS COCO 2017 dataset [23] for object detection, demonstrate that CADFC MobileNet achieves to enhance comprehensive strength.

The main contributions of this paper can be summarized as follows:

(1) A new context-aware Decoupled Fully Connected (CADFC) attention mechanism is presented. This method use dual representation, i.e., remote context and local context, to construct remote and local attention maps, can almost possess a global perspective.
(2) A new CADFC bottleneck block is introduced, which aggregates two feature branches: the depthwise convolution branch focuses on extracting spatial features from localized regions, while the CADFC branch captures global features based on these focused regions. This design allows for flexible extraction of key information from the input and efficient communication of this information to the head component.
(3) A new CADFC MobileNet is proposed as visual backbone network. This network has optimized the mainstream MobileNetV2 architecture for edge devices, ensuring strong overfull performance on image classification and object detection tasks while maintaining efficiency.

2. Related work

In this section, we elaborate on mainstream lightweight visual neural networks and attention mechanism.

2.1. Lightweight networks

2.1.1. CNN-based lightweight networks

In order to reduce computational load and model size while maintaining accuracy, many lightweight CNN networks have proposed different optimization strategies. These can be categorized based on their optimization objectives and application scenarios.

Networks Optimized for Parameters and Computational Load. These networks typically reduce the computational burden through structural innovations such as decomposed convolutions and channel rearrangements. MobileNet Series [15,24] are used as backbone networks, which introduces depthwise separable convolution, which splits the standard convolution operation into depthwise convolution and pointwise convolution, significantly reducing computational load. ShuffleNet Series [16,25] are used as backbone networks, which uses channel shuffling to promote information exchange between different channel groups, improving information flow while reducing computational cost. SqueezeNet [26,27] are mainly used for classification tasks, which introduces the Fire module (composed of 1×1 and 3×3 convolutions) to reduce the number of convolution filters, significantly lowering the model's parameter count.

Networks Focused on Enhancing Feature Extraction and Information Flow. This category focuses on improving feature extraction capabilities and enhancing information flow while reducing computational load. These networks typically introduce mechanisms such as feature selection, attention mechanisms, and frequency domain optimization to improve performance in complex visual tasks. They have been developed using various techniques to reduce the number of parameters and computational complexity while maintaining high accuracy. GhostNet [28] introduces the Ghost module to generate "virtual" features as replacements for conventional convolutions, GhostnetV2 [10] builds on GhostNet by incorporating a Dual Feature Consistency (DFC) attention mechanism to optimize feature selection and enhance global information flow. It is used as a backbone network for classification tasks, improving global context perception. AFFNet [29] consisting of adaptive frequency filters (AFF), which is proposed as a lightweight visual backbone network, introduces AFF Token Mixers and adaptive frequency filters (AFF) to extract global information efficiently.

Besides manual design, there are a series of models constructed by the Neural Architecture Search (NAS) framework, such as ProxylessNet [30], MNASNet [31], and FBNet [32]. ProxylessNet is an efficient neural network architecture that employs neural architecture search (NAS) techniques to identify the optimal network structure, reducing the need for manual design. MNASNet utilizes reinforcement learning to automatically search for network architectures, thereby finding the most suitable model structure for specific tasks. FBNet involves a flexible search space, which explores different combinations of operations to find the optimal architecture, ensuring adaptability to specific hardware platforms and task requirements.

Recently, StarNet [33] enhances feature expressiveness and information flow by projecting features into a higher-dimensional implicit feature space through the Star operation, thereby improving feature representation. This approach focuses on adaptive and high-dimensional feature space optimization, allowing the network to better capture complex feature relationships, which in turn boosts performance. WTConv [34] is introduced primarily to effectively increase the receptive field of convolutions. By incorporating Wavelet Transform Convolutions (WTConv), it expands the receptive field of convolutional layers, helping the network capture a larger range of contextual information. One advantage of this method is that it can serve as a "plug-and-play" replacement in existing architectures, enhancing perceptual capabilities. In object detection tasks, besides the classic YOLO series, many lightweight networks have been proposed in recent years to replace YOLO's functionality, better meeting the efficiency requirements of edge devices. For instance, DPNet [35] introduces a dual-path network that uses a lightweight attention mechanism to extract high-level semantic features and low-level detail features in parallel, thereby improving object detection accuracy, especially in resource-constrained environments.

2.1.2. Transformer-based lightweight networks

DeiT [11] uses the token-based distillation method to enable transformer-based visual models with low parameter counts to achieve comparable accuracy to convolutional models on small datasets. B. Heo et al. [36] combine Patchwise Transformer (PiT) and knowledge distillation (Distillation) techniques to provide an effective solution that reduces model size and computational complexity while maintaining performance. L.Yuan et al. [37] proposed a Token-to-Token Vision Transformer model (T2T ViT) that has been processed through knowledge distillation, featuring a specific scale or depth design. Pyramid Vision Transformer (PVT) [38] improves traditional Vision Transformers by incorporating a pyramid structure, efficient attention mechanisms, and hierarchical feature representations.

2.1.3. Hybrid lightweight networks

Hybrid lightweight neural networks, such as MobileViT [39] combines a convolutional backbone with a transformer-based head that utilizes self-attention mechanisms to capture global dependencies and achieves state-of-the-art performance on several benchmark datasets

while maintaining high computational efficiency. CeiT-T [40] leverages the local feature extraction capability of convolutional layers to capture local features and texture information, while employing a self-attention mechanism in the transformer to handle global dependencies. LocalViT-T [41] combines the capability of local feature extraction using convolutional layers with the local attention mechanism and the global modeling ability of vision transformers, allowing it to extract low-level detail features and high-level semantic features.

2.2. Attention mechanism

The Transformer architecture, particularly with its core Multi-Head Attention mechanism [42], can capture global context, has significant advantages when handling long-sequence data. However, Multi-Head Attention requires storing a large number of intermediate computation results and attention matrices, the computational and memory consumption can become a bottleneck, especially on resource-constrained devices. Therefore, in this study, we focus on the attention mechanism in CNN. CNN effectively captures local features through convolution operations, and progressively extracts features from local to global through multi-layer structures such as convolutional layers and pooling layers. There are two main types of adding attention to CNNs.

The first category emphasizes important feature channels and positions, known as channel-wise and spatial-wise attention, learned through the network. SENet [43] uses pooling operations to encode global context. GSNet [44] explores second-order global pooling to reweight feature channels. In addition to channel attention, some networks [45,46] focus on learning spatial attention to capture global positional context. For instance, CBAM [45] applies average and max pooling in the channel and spatial dimensions, respectively. ECANet [47], the applications of this approach [48,49], replaces fully connected layers with 1-D convolutional layers. Although capturing context through global pooling is computationally efficient, it remains weak in representing element-wise interactions.

The second category captures global context by calculating correlation matrices between each image element. The Non-local network [50], a pioneer, models pixel-to-pixel relationships, where each position is reweighted by all other positions. GCNet [51] suggests that only reweighting feature channels is necessary to capture global context. Dual Feature Consistency (DFC) [10] is applied to GhostNetV2 to enhance feature consistency in the network. It aims to capture both local and global dependencies within an image by measuring the relationships between different features or spatial elements in the image.

To address the limitations of resource-constrained edge devices, our approach focuses on optimizing a lightweight backbone network with a novel attention mechanism in CNN. By revisiting the backbone network MobileNetV2 and incorporating global perception mechanisms into depthwise separable convolutions, we uses two low-rank convolutions of per channel to process input features in parallel, aim to preserve local feature representations while enhancing the model's ability to capture global context. This strategy ensures improved accuracy across multiple tasks, including classification and detection. This approach is particularly essential for edge devices such as smartphones, embedded systems, and IoT devices, which often need to perform a variety of related tasks simultaneously. Training and optimizing separate models for each task is impractical. By leveraging a shared lightweight backbone and integrating global perception mechanisms, features can be efficiently shared across tasks, leading to improved performance and mitigating the need to deploy and train individual models for each task.

3. Preliminary

3.1. Backbone: MobileNetV2

MobileNetV2 [24] is a lightweight deep learning model specifically tailored for resource-constrained mobile devices. It has gained widespread adoption and achieved outstanding performance in various real-world applications. At the core of MobileNetV2 lies the MobileNetV2 module, which features an inverted residual structure. The module's computations can be decomposed into three steps. Given the input feature tensor X, where H is the height, W is the width, and C_{in} is the number of input channels, the MobileNetV2 module first employs pointwise convolution to generate redundant feature maps,

$$Y' = X^{H\times W\times C_{in}} * \mathrm{Conv}_{1\times 1} \tag{1}$$

where $*$ denotes the convolution operation. $Y' \in R^{H\times W\times C'}$ represents the redundant feature map engendered through pointwise convolution. Typically, the number of output channels satisfies $C' > C_{in}$. The redundant feature map preserves richer feature information. Leading to better feature representation after the subsequent depthwise convolution,

$$Y'' = Y' * \mathrm{DWConv}_{3\times 3} \tag{2}$$

Where $Y'' \in R^{H\times W\times C'}$. To compensate for the lack of information in the channel dimension of depthwise convolution, another pointwise convolution is performed to aggregate information in the channel direction and reduce it to match the input channel number. Finally, through residual connection, the output of the convolution is added to the input features to avoid information loss and improve the stability of the model training,

$$Y = ADD\left(\left[X, Y'' * \mathrm{Conv}_{1\times 1}\right]\right) \tag{3}$$

$Y \in R^{H\times W\times C_{out}}$ is the output feature. MobileNetV2 is constructed by stacking multiple bottlenecks, which are augmented with design strategies such as linear bottlenecks and SE [43] modules to further boost the network's efficiency and accuracy. However, due to the inherent limitation of convolutions in capturing global feature representations, MobilenetV2 has limited capability in extracting global features.

3.2. Feature aggregation: DFC module

The core idea of DFC [10] is to use fully connected layers to replace self-attention to obtain the attention map of the global image and reduce computational complexity during the process of obtaining global information. Specifically, given a feature map $Y \in R^{H\times W\times C}$, where HW can be seen as tokens $z_i \in R^c$, the attention map $A = \{a_{11}, a_{12,\ldots,} a_{HW}\}$ is calculated as follows:

$$a_{hw} = \sum_{h',w'} F_{hw,h'w'} \odot z_{h'w'} \tag{4}$$

Where $\odot$ denotes element-wise multiplication, and F is a learnable weight of the fully connected layer. For example, given $Z = \{z_{11}, z_{12,\ldots,} z_{HW}\}$, we can obtain the attention map $A = \{a_{11}, a_{12,\ldots,} a_{HW}\}$. However, the computational complexity of the attention map is still quadratic $\left(\mathcal{O}\left(H^2W^2\right)\right)$. To further reduce computational costs, DFC constructs long-range relationships between pixels in the horizontal and vertical directions of the feature map through two low-rank convolutions, which ensures efficient computation while obtaining a global feature representation of the feature map. The calculation is as follows:

$$a''_{hw} = \sum_{h'=1}^{H} F^{H}_{h,h'w} \odot z_{h'w}, h = 1, 2, \ldots, H, w = 1, 2, \ldots, W, \tag{5}$$

$$a'_{hw} = \sum_{w'=1}^{W} F^{W}_{w,hw'} \odot a'_{hw'}, h = 1, 2, \ldots, H, w = 1, 2, \ldots, W, \tag{6}$$

DFC uses two low-rank convolutions with kernel sizes of $1 \times K_H$ and $K_W \times 1$ to calculate conversion weights F^H and F^W in sequence. After replacing the fully connected layer with convolution, the computational complexity theoretically reduces to $\mathcal{O}\left(K_H HW + K_W HW\right)$. However, as shown in Fig. 1, the attention maps constructed by DFC in the early layers of the network (i.e., Layer 5) exhibit a clear cross pattern, indicating that it focuses more on the horizontal and vertical correlations between elements during weight learning, which tends to lose information in the corners of the feature map.

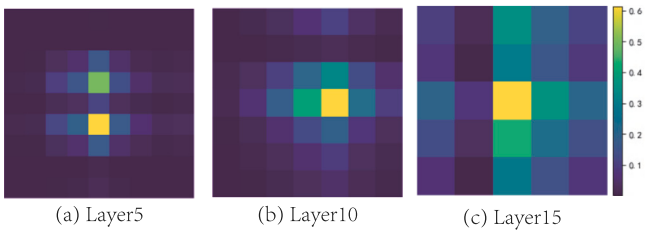


Fig. 1. Visualization of DFC attention map from different layers.

4. Approach

In this section, we will describe the process of constructing a context-aware DFC attention mechanism at the bottom level and gradually optimizing it to develop a new visual mobile network.

4.1. Context-aware DFC attention

To enhance the incorporation of global information, we propose a new *context-aware DFC attention*, called CADFC, to build the attention maps. Fig. 2 illustrates the comprehensive process involved in the generation of these attention maps. Given an input feature map $X \in R^{H\times W\times C}$ with dimensions H (height), W (width), and C (number of channels), it is divided into N non-overlapping uniform patches, each sized $h \times w$. The feature map is reshaped into $X_U \in \mathbb{R}^{P\times N\times C}$. The dimension P represents the number of elements contained in each patch, the dimension N indicates the number of patches, and the third dimension C remains the number of channels.

The main idea of CADFC lies in decomposing attention into two aspects: remote context-aware attention and local context-aware attention. On the one hand, remote context-aware attention is designed to extract relationships between the elements across different patches, i.e., remote context, which is reflected in the processing of vertical attention maps. On the other hand, local context-aware attention focuses on the relationships among elements within a single patch, i.e., local context, contributing to the horizontal attention map processing.

When processing in the vertical direction, learnable weights in the convolutional layer establish sparse connections with surrounding points across various spatial positions. Each point connects to corresponding points in all other patches to calculate their correlations, resulting in a receptive field for CADFC in the vertical direction that spans is $H \times W$. Given F as the learnable weight parameters, with $H = nh$ and $W = nw$, where $h \times w$ is the size of patches. For each position (i, j) in the corresponding position in the patch, the vertical attention map for each position (i, j) in the corresponding patch is defined as

$$a_{i,j} = \sum_{l=0}^{h-1}\sum_{k=0}^{w-1} f_{l,k} \odot y_{l*h+i,k*w+j} \tag{7}$$

When the convolution builds relationships between elements in the horizontal direction, the elements involved in the calculation are the pixels within each patch. The horizontal attention map for each position (i, j) is defined as:

$$a_{i,j} = \sum_{l=0}^{h-1}\sum_{k=0}^{w-1} f_{l,k} \odot y_{l,k} \tag{8}$$

Eqs. (7) and (8) explain how CADFC aggregates remote and local context in the feature map to construct attention maps. Compared to the original DFC method, CADFC not only builds long-range remote context but also aggregates more local context within each patch. By combining remote and local context, c-DFC can construct attention maps that are closer to full attention. The complete algorithm is shown in the (c) section of Fig. 3.

4.2. CADFC bottleneck block

Fig. 3 shows the design paradigm of the CADFC Bottleneck component. For the same set of feature maps, the CADFC module and depthwise convolution module extract features with different reception fields simultaneously. CADFC module establishes long-range dependencies through its own advantages to aggregate remote context cross patches and local context in patches. The depthwise convolution module has the inductive bias characteristic, which allows it to effectively capture local feature information. The combination of the two parts in a fusion module enables the final output features to obtain rich global information and local expressive ability.

Specifically, given a set of input tensors $X \in \mathbb{R}^{H\times W\times C}$, it is copied and send to two different branches, respectively. The fusion module is used to aggregate information from the two branches shown in Fig. 3(b). The depthwise convolution branch is responsible for extracting local features. The CADFC branch is responsible for extracting global features. To ensure the high efficiency of network computing and obtain multi-scale reception field feature maps, one of them is processed by 3×3 depthwise convolution to obtain the output tensor Y. The other parts are processed by CADFC to obtain the attention map A. As illustrated in Fig. 3(c), CADFC unfolds the downsampled features and applies two low-rank convolutions to extract information from the input features. Subsequently, it reconstructs the image by performing upsampling through nearest-neighbor interpolation, generating a feature map that aligns with the dimensions of the input features. Finally, the feature maps for two branches are multiplied element by element to obtain the output feature $Z \in \mathbb{R}^{H\times W\times C}$. The formula is:

$$Z = Y \odot Sigmoid\,(A) \tag{9}$$

Where $\odot$ represents element-wise multiplication. we also apply the $Sigmoid$ function on the attention map to normalize it to the range of $(0, 1)$, same as the original settings in DFC.

We also introduce the Squeeze-and-Excitation (SE) module [43] in the component to aggregate more channel-wise information and improve the representation ability of features. Finally, a residual connection is used on the input feature map and output feature map to speed up network convergence [20].

4.3. CADFC MobileNet

The reliability of MobileNetV2 has been verified in many practices, so we choose the MobileNetV2 as the basis for implementing the backbone network. According to the empirical study of MobileNetV2, we use standard convolution with a stride of 2 in the first step to generate basic feature maps with relatively intact information. Moreover, setting the expansion rate value that controls the channel dimension expansion to 6 can achieve the best network performance. However, greatly expanding the number of feature channels will increase the latency of network inference. Therefore, we only use the recommended expansion rate value in the latter half of the network and choose smaller values for the rest, Specific network details are shown in Table 1.

Although the computation of the CADFC attention map is efficient, it still introduces additional computational complexity. To address this, we adopt a design principle consistent with [10], downsampling the input features by half. By applying CADFC attention to the downsampled features, we effectively reduce the computational complexity to one-quarter of the original. To ensure that the attention map aligns in size with the corresponding feature map, we utilize bilinear interpolation for upsampling the attention map. Furthermore, to mitigate the computational latency associated with the $sigmoid$ function, we selectively apply it only to the downsampled attention map. This strategy minimizes the number of parameters processed by the activation function, thereby reducing overall computational costs and enhancing practical latency. Additionally, to accommodate varying application requirements, the CADFC MobileNet allows for width adjustments, facilitating a better trade-off between accuracy and efficiency.

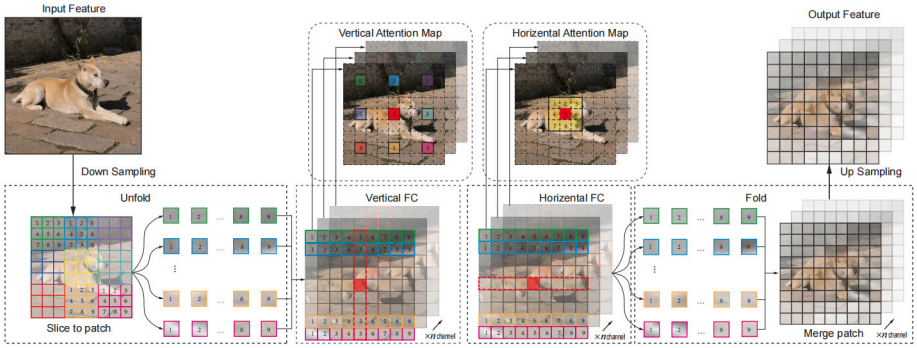
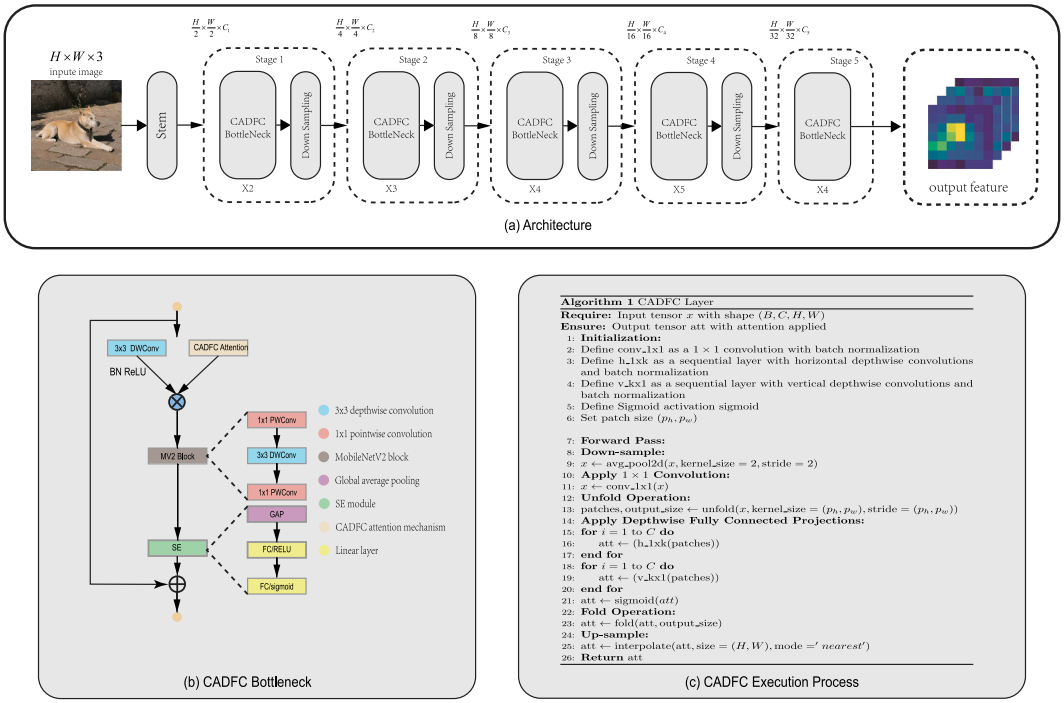


Fig. 2. Illustration of the information flow of CADFC attention. The horizontal and vertical fully connected layers are instrumental in modeling the long-range dependencies between patches and the local dependencies within patches, respectively.



Algorithm 1 CADFC Layer

Require: Input tensor x with shape (B, C, H, W)
Ensure: Output tensor att with attention applied

Initialization:
Define conv_1x1 as a 1×1 convolution with batch normalization
Define h_1xk as a sequential layer with horizontal depthwise convolutions and batch normalization
Define v_kx1 as a sequential layer with vertical depthwise convolutions and batch normalization
Define Sigmoid activation sigmoid
Set patch size (p_h, p_w)

Forward Pass:
Down-sample:
$x \leftarrow \text{avg_pool2d}(x, \text{kernel_size} = 2, \text{stride} = 2)$
Apply 1×1 **Convolution:**
$x \leftarrow \text{conv_1x1}(x)$
Unfold Operation:
$\text{patches}, \text{output_size} \leftarrow \text{unfold}(x, \text{kernel_size} = (p_h, p_w), \text{stride} = (p_h, p_w))$
Apply Depthwise Fully Connected Projections:
for $i = 1$ to C **do**
    $\text{att} \leftarrow (\text{h_1xk}(\text{patches}))$
end for
for $i = 1$ to C **do**
    $\text{att} \leftarrow (\text{v_kx1}(\text{patches}))$
end for
$\text{att} \leftarrow \text{sigmoid}(att)$
Fold Operation:
$\text{att} \leftarrow \text{fold}(\text{att}, \text{output_size})$
Up-sample:
$\text{att} \leftarrow \text{interpolate}(\text{att}, \text{size} = (H, W), \text{mode} = ' nearest')$
Return att

(c) CADFC Execution Process

Fig. 3. (a) CADFC MobileNet architecture detailed design. (b) The process of the CADFC bottleneck, (c) CADFC execution process.

5. Experiments

In this section, we present a comprehensive empirical study and performance evaluation of the proposed CADFC MobileNet across two fundamental computer vision tasks: image classification and object detection Our experiments aim to assess the overall efficacy of the network in addressing these tasks. Additionally, we include the results of an ablation study at the conclusion of this section to provide deeper insights into the proposed network's capabilities. The experiments were conducted using PyTorch [52] on a Linux operating system. The software environment was configured with CUDA 11.8, cuDNN 7.6, and PyTorch version 3.10. The hardware utilized for these experiments included an Intel Xeon 4110 CPU and six NVIDIA A100 PCIE 40G GPUs.

Warmup training is a common technique used to adjust the learning rate strategy. During the initial stages of training, when the model weights are randomly initialized, there is a possibility of encountering local optima. The Warmup strategy mitigates the risk of the model getting trapped in local optima by using a smaller learning rate for the first few epochs. Additionally, in the early stages of training, when the model parameters have not yet been sufficiently adjusted, training instability may occur. Warmup strategies help stabilize the training process, preventing issues such as gradient explosion or gradient vanishing. Therefore, we incorporated the warmup strategy in all subsequent experiments.

5.1. Image classification

Committed to image classification tasks, we evaluated our network on the ImageNet 2012 dataset [7], which is a large-scale dataset with 1000 classes, consisting of 1 million training images and 50,000 testing images. All images were designed to have a size of 224 × 224 pixels. The accuracy metric is typically defined as the proportion of correctly predicted instances (both True Positives TP and True Negatives TN) among the total instances (both Total Positives P and Total Negatives N). The formula for accuracy is:

$$Accuracy = \frac{TP + TN}{P + N} \tag{10}$$

Top-1 and Top-5 accuracy are commonly used as evaluation metrics for model performance. Top-1 accuracy refers to the proportion of predictions with the highest probability that match the actual labels. In Top-5 accuracy, the model provides the top five predictions with the highest probabilities, and if any of these predictions match the actual label, the classification is considered correct. Moreover, FLOPs (floating-point

Table 1
CADFC MobileNet architecture: Each line describes a sequence of 1 layer. "t" represents the expansion factor in MobileNetV2 block, which is the multiplier for the number of channels in each layer, and "c" represents the number of output channels. Except for the layers that require downsampling with a stride of "s", the stride for all other layers is set to 1. In the SE block, "reduction ratio" (or simply "ratio") represents the reduction factor applied to the number of channels during the squeeze operation. A reduction ratio of 0 indicates that the SE block is not used, meaning no channel reduction is performed.

Layer	Input	Operator	t	c	s	se
1	$224^2 \times 3$	Conv2d 3 × 3	–	24	2	0
2	$112^2 \times 24$	Bottleneck	2	32	1	0
3	$112^2 \times 32$	Bottleneck	3	40	2	0
4	$56^2 \times 40$	Bottleneck	3	40	1	0
5	$56^2 \times 40$	Bottleneck	3	40	1	0
6	$56^2 \times 40$	Bottleneck	3	64	2	0.25
7	$28^2 \times 64$	Bottleneck	3	64	1	0.25
8	$28^2 \times 64$	Bottleneck	3	64	1	0
9	$28^2 \times 64$	Bottleneck	3	64	1	0
10	$28^2 \times 64$	Bottleneck	4	96	2	0
11	$14^2 \times 96$	Bottleneck	4	96	1	0
12	$14^2 \times 96$	Bottleneck	4	96	1	0
13	$14^2 \times 96$	Bottleneck	4	128	1	0
14	$14^2 \times 128$	Bottleneck	4	128	1	0
15	$14^2 \times 128$	Bottleneck	6	192	2	0.25
16	$7^2 \times 192$	Bottleneck	6	192	1	0
17	$7^2 \times 192$	Bottleneck	6	192	1	0.25
18	$7^2 \times 192$	Bottleneck	6	192	1	0
19	$7^2 \times 192$	Bottleneck	6	192	1	0.25
20	$7^2 \times 192$	AvgPool 7 × 7	–	192	–	–
21	$1^2 \times 1280$	Conv2d 1 × 1	–	1280	–	–
22	$1^2 \times 1280$	FC	–	1280	–	–

operations) are used to measure the computational complexity of a model and typically represent the amount of computation required during model inference.

Settings According to the recommended hyperparameter settings provided by GhostNetV2 [10], the Root Mean Square Propagation (Rmsprop) optimizer is used to update the model weights. During the warmup phase of training, the learning rate is initially set to 0.000001. After the warmup phase, the learning rate is increased to 0.64. Additionally, a step schedule learning rate decay strategy is employed, where the learning rate is reduced by a factor of 0.973 every 2.4 epochs. The remaining hyperparameter settings are as follows: the weight decay coefficient is set to 0.00001, the dropout rate is set to 0.3, and the model's exponential moving average decay is set to 0.9999. The model is trained for 450 epochs on the ImageNet dataset, and to maintain consistency with the reference settings, a batch size of 1026 is used.

Results and Analysis Table 2 shows the performance of CADFC MobileNet and other state-of-the-art neural networks on the ImageNet dataset. Compared to the state-of-the-art lightweight CNNs, including the widely used MobileNet series [15,24,53], ShuffleNet series [16,25], and recent SOTA network GhostNetV2 [10], StarNet [33], and WTConv [34].

A notable advantage of CADFC MobileNet is its ability to achieve relatively high accuracy while maintaining a relatively small number of parameters. Compared to other mainstream lightweight networks such as GhostNetV2, StarNet, and MobileViT, CADFC MobileNet1.0X achieves a Top-1 accuracy of 76.8%, surpassing GhostNetV2's 75.3% and StarNet-S1's 73.5%. This result indicates that CADFC MobileNet enhances the model's expressive power by optimizing the network structure and reducing redundant parameters. CADFC MobileNet provides a favorable balance between low computational complexity (fewer FLOPs) and high accuracy, outperforming other networks in this regard. Even though networks like WTConvNeXt-T may outperform CADFC MobileNet in accuracy (by around 7.7%), they have approximately 5.4 times more parameters, making CADFC MobileNet a more efficient choice for practical deployment. In comparison to other transformer-based or hybrid models like MobileViT-XS (74.8% accuracy), CADFC MobileNet0.9X achieves 76.1% with significantly lower computational costs, further underscoring its efficiency in real-world applications.

The CADFC mechanism in the Bottleneck block helps the model better associate different parts of the input. With the attention mechanism, the network can globally adjust the importance of each input feature, assigning different weights to specific parts of the input at each step. This dynamic weighting enhances the model's ability to focus on the most relevant features, which is particularly beneficial for image classification tasks. By considering global context and adapting attention based on the content, the model can more effectively distinguish between different image features, leading to improved classification performance.

5.2. Object detection

we assessed the performance of our network in object detection tasks on the MS COCO 2017 dataset [23], which comprises 118,000 training images and 5000 validation images. The evaluation metrics for object detection tasks include AP, AP_{50}, AP_{75}, AP_S, AP_M, and AP_L. AP is a critical metric for measuring the effectiveness of object detection algorithms, calculated as the average precision across various recall levels. Specifically, AP_{50} and AP_{75} represent the average precision at Intersection over Union (IoU) thresholds of 0.50 and 0.75, respectively. AP_S, AP_M, and AP_L correspond to the average precision for small, medium, and large objects, respectively, allowing for a detailed analysis of the model's performance across different object sizes.

Settings We evaluated the performance of CADFC MobileNet for object detection tasks by integrating it into the lightweight YOLOv3 model [22] and training it on the MS COCO 2017 dataset. The detection head used in GHOSTNETV2 [10] is based on the YOLOv3 version, so you should select the YOLOv3 detection head for consistency. This ensures compatibility and aligns with the architecture that GHOSTNETV2 employs. YOLOv3 is renowned for being one of the fastest one-stage object detection algorithms, and despite the emergence of newer models, YOLOv3 remains widely adopted in state-of-the-art methods for its efficiency [55–57].

For the training process, we utilized pre-trained weights from ImageNet and fine-tuned them for object detection. The training setup employed Stochastic Gradient Descent (SGD) as the optimizer for 100 epochs, with a batch size of 512. Initially, the learning rate was set to 0.003, and after three warmup epochs, it was increased to 0.01. Other hyperparameters followed the guidelines provided by MMDetection,[1] with the momentum parameter set to 0.937, warmup momentum to 0.8, and weight decay to 0.0005.

To enhance the COCO dataset images and improve model robustness, we applied the mosaic data augmentation technique, which includes operations such as cropping, rotation, brightness adjustment, and noise addition. Since COCO images come in varying sizes, we resized them to standard input resolutions of 320 × 320 and 416 × 416, and all experimental results were based on these input sizes.

Result and Analysis Table 3 presents a performance comparison of the proposed CADFC MobileNet against other networks on the MS COCO dataset. Based on YOLOv3, the model using CADFC MobileNet 1.0X as the backbone network performs exceptionally well. For a resolution of 320 × 320, the mAP is 29.0, and the AP50 is 47.7. At a higher resolution of 416 × 416, the mAP is 35.1, and the AP50 is 55.1, both of which are the highest values in the YOLOv3-based cases of the table. This indicates that this configuration provides significant improvements for small to medium-sized objects, especially at higher resolutions, where the accuracy shows a marked enhancement.

[1] https://github.com/open-mmlab/mmdetection

Table 2
Accuracies on ImageNet and ImageNet real of CADFC MobileNet and of several SOTA CNNs, Transformers and Hybrid models, for models trained with no extra data.

Type	Model	Params (M)	FLOPS (M)	Top-1 Acc.	Top-5 Acc. (%)
Transformers	DeiT-T [11]	5.7	1300	72.2	91.1
	DeiT-S [11]	22.1	4500	79.9	95.0
Transformers	PiT-Ti+Distill [36]	4.9	700	74.6	–
	PiT-XS+Distill [36]	10.6	1400	79.1	–
Transformers	T2T-ViT-7-Distilled [37]	4.3	1100	73.1	–
	T2T-ViT-12-Distilled [37]	6.9	1900	77.4	–
Transformers	PVT-Tiny [38]	13.2	1900	75.1	–
	PVT-Small [38]	24.5	3800	79.8	–
Hybrid	MOBILEVIT-XS [39]	2.3	700	74.8	–
	MOBILEVIT-S [39]	5.6	2000	78.4	–
Hybrid	CeiT-T [40]	6.4	1200	76.4	93.4
	CeiT-S [40]	24.2	4500	82.0	95.9
Hybrid	LocalViT-T [41]	5.9	1300	74.8	92.6
	LocalViT-S [41]	22.4	4600	80.8	95.4
Hybrid	ConViT-Ti [54]	6.0	1000	73.1	91.7
	ConViT-Ti+ [54]	10.0	2000	76.7	93.6
CNNs	MOBILENETV2 0.6X [24]	2.2	141	66.7	–
	MOBILENETV2 1.0X [24]	3.5	300	72.0	90.8
CNNs	MOBILENETV3-L 0.75X [53]	4.0	155	73.3	–
	MOBLIENETV3-L 1.0X [53]	5.4	219	75.2	–
	MOBILENETV3-L 1.25X [53]	7.5	355	76.6	–
CNNs	SHUFFLENETV2 1.0X [25]	2.3	146	69.4	88.9
	SHUFFLENETV2 1.5X [25]	3.5	299	72.6	90.6
CNNs	GHOSTNETV1 1.0X [28]	5.2	141	73.9	91.4
	GHOSTNETV1 1.3X [28]	7.3	226	75.7	92.7
	GHOSTNETV1 1.7X [28]	11.0	378	77.2	93.4
CNNs	GHOSTNETV2 1.0X [10]	6.1	167	75.3	92.4
	GHOSTNETV2 1.3X [10]	8.9	269	76.9	93.4
	GHOSTNETV2 1.6X [10]	12.3	399	77.8	93.8
CNNs	CADFC MobileNet0.9X (Ours)	**4.7**	**532**	**76.1**	**93.0**
	CADFC MobileNet1.0X (Ours)	**5.2** (−0.9)	**592**	**76.8** (+1.5)	**93.1** (+0.7)
	CADFC MobileNet1.3X (Ours)	**7.6** (−1.3)	**986**	**78.5** (+1.6)	**94.1** (+0.7)
	CADFC MobileNet1.6X (Ours)	**10.6** (−1.7)	**1405**	**79.5** (+1.7)	**94.3** (+0.5)
CNNs	PROXYLESSNAS [30]	4.1	320	74.6	92.2
CNNs	MNASNET-A1 [31]	3.9	312	75.2	92.5
	MNASNET-A2 [31]	4.8	340	75.6	92.7
	MNASNET-A3 [31]	5.2	403	76.7	93.3
CNNs	FBNET-B [32]	4.5	295	74.1	–
	FBNET-C [32]	5.5	375	74.9	–
CNNs	StarNet-S1 [33]	2.9	425	73.5	–
	StarNet-S2 [33]	3.7	547	74.8	–
	StarNet-S3 [33]	5.8	757	77.3	–
	StarNet-S4 [33]	7.5	1075	78.4	–
CNNs	WTConvNeXt-T [34]	30	4500	82.5	–
	WTConvNeXt-S [34]	54	8800	83.6	–
	WTConvNeXt-B [34]	93	15500	84.1	–

The results indicate that CADFC MobileNet significantly improves the recognition of large, medium, and small objects when compared to previous networks. Specifically, relative to the advanced GHOSTNET V2.1.0X with a resolution of 416 × 416, CADFC MobileNet achieves a 14.2% increase in Average Precision (AP) for large objects, reaching 50.3%. It also improves the AP for medium-sized objects by 13.4 percentage points, resulting in a performance of 38.4%. In terms of small object detection, the AP rises from 10.4% to 17.3%. The attention mechanism in CADFC MobileNet allows the model to dynamically focus on the most relevant features in the image, which is especially useful in complex environments with distractions or occlusions, such as the bear camouflaged in trees 4 (a), birds of varying sizes 4, or obstructed motorcycles 4 . Here's how the attention mechanism aids in these tasks: The CADFC attention mechanism, particularly when integrated with the Nest module, enables the model to focus selectively on salient features (such as the bear, motorcycles, or the person in the scene) while ignoring irrelevant background elements (like trees or complex textures). Thereby significantly enhancing the accuracy of object recognition across various sizes. Furthermore, this capability allows the model to concentrate on details pertinent to the target within complex scenes, ensuring efficient detection and localization in diverse environmental conditions. These findings indicate that CADFC MobileNet exhibits exceptional flexibility, making it well-suited for detecting a wide range of objects.

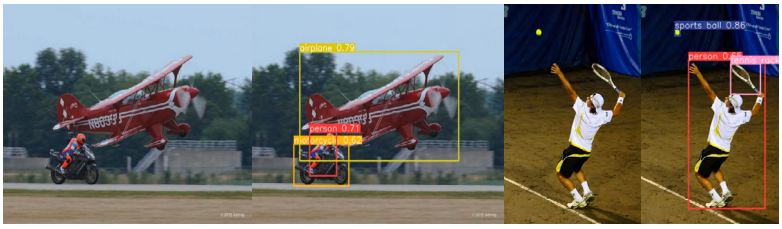
Moreover, DPNet performs well at a resolution of 320 × 320, achieving an mAP of 29.6. Although there is no specific data for large or small objects, the overall mAP still shows good performance. PP-YOLO and PP-YOLOv2 both use ResNet50-vd-dcn as the backbone network and perform excellently, particularly at higher resolutions. At 416 × 416 resolution, PP-YOLO achieves an mAP of 42.5 and an AP50 of 62.8, which is already very high compared to other models. PP-YOLOv2 further improves upon this performance. At 416 × 416 resolution, it achieves an mAP of 46.3 and an AP50 of 65.1, with particularly strong performance on large objects (AP_L), reaching 62.2. Under resource constraints, PP-YOLOv2, CADFC MobileNet1.0X, and DF are all excellent



(a) Single-target simple scene



(b)Multiple targets of a single category



(c) Multiple targets of different categories

Fig. 4. Detection results in different scenarios.

Table 3
Results of object detection on MS COCO dataset.

Model	Backbone	Resolution	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Tiny YOLOv3 [22]	Tiny-DarkNet	416 × 416	16	33.1	–	–	–	–
YOLOv3 [22]	DarkNet-53	320 × 320	28.2	51.5	29.7	–	–	–
Tiny YOLOv4 [58]	CSPDarkNet53-Tiny	416 × 416	21.7	40.2	–	–	–	–
YOLOv3	MOBILENET V2 1.0X	320 × 320 416 × 416	22.2 23.9	41.9 45.4	21.4 22.6	6.0 10.6	23.6 25.1	35.8 34.9
YOLOv3	GHOSTNET V1 1.1X	320 × 320 416 × 416	21.8 23.4	41.2 45.2	20.8 21.9	5.7 9.8	22.3 24.4	37.3 34.9
YOLOv3	GHOSTNET V2 1.0X	320 × 320 416 × 416	22.3 24.1	41.4 45.7	21.9 23.0	6.0 10.4	22.8 25.0	38.1 36.1
YOLOv3	CADFC MobileNet1.0X	320 × 320 416 × 416	**29.0** **35.1**	**47.7** **55.1**	**29.9** **37.0**	**13.2** **17.3**	**31.2** **38.4**	**43.2** **50.3**
Lightweight detection head	DPNet [35]	320 × 320	29.6	46.9	31.3	–	–	–
PP-YOLO [59]	ResNet50-vd-dcn	320 × 320 416 × 416	39.3 42.5	59.3 62.8	42.7 46.5	16.7 21.2	41.4 45.2	57.8 58.2
PP-YOLOv2 [60]	ResNet50-vd-dcn	320 × 320 416 × 416	43.1 46.3	61.7 65.1	46.5 50.3	19.7 23.9	46.3 50.2	61.8 62.2

choices. However, if the goal is to minimize configuration overhead while also addressing object classification tasks, CADFC MobileNet1.0X is more suitable. After completing the classification task, you can easily switch to the YOLOv3 detection head for object detection tasks.

5.3. Ablation studies

In this section, we conducted a comprehensive empirical study to analyze the impact of each component on CADFC MobileNetNet. All experimental results were obtained by training CADFC MobileNet1.0X on ImageNet under the same training configuration on 6 A100 40G GPUs.

The impact of the proposed key component. To evaluate the effectiveness of the proposed CADFC, we conduct a series of ablation experiments, as summarized in Table 4. In these experiments, we compare CADFC with several classic and state-of-the-art (SOTA) attention mechanisms, including CBAM [45], DFC [10], and Multi-Head Self-Attention (MHSA) [42]. The results demonstrate that, at epoch 100, the classification network incorporating CADFC achieves higher accuracy than the other attention modules. Specifically, CADFC achieves a top-1 accuracy of 69.69%, surpassing CBAM (68.91%), DFC (69.09%), and MHSA (67.57%) in final accuracy. In terms of model size, CADFC has 5.26M parameters, which is comparable to CBAM (5.06M) and DFC (5.16M), and slightly fewer than MHSA (5.86M). These results indicate that CADFC strikes the best balance between accuracy and computational efficiency. It achieves the highest top-1 accuracy among the attention mechanisms tested, with only a marginal increase in the number of parameters.

The impact of patch size in CADFC attention. CADFC MobileNet bottleneck combines the global image information extracted by CADFC and the local information extracted by MV2. In CADFC, the feature map is partitioned into N non-overlapping patches with height and

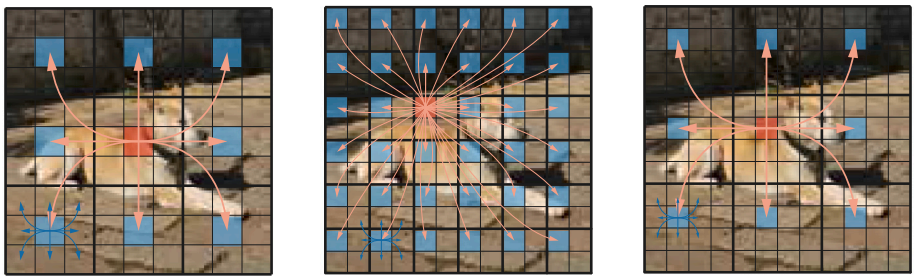


Fig. 5. Relationship between kernel size ($n \times n$) for convolutions and patch size ($h \times w$) for folding and unfolding in CADFC. By incorporating both local (blue-colored arrows) and global (orange-colored arrows) information, the red pixel in (a) and (b) can aggregate information from all pixels. In contrast, in (c), the red pixel can only aggregate local information within each patch, resulting in information loss. The grids segmented by bold black lines and thin black lines represent patches and pixels, respectively.

Table 4
Ablation experiment results under different attention mechanisms.

Attention modules	# Params	Top-1 ACC. (%)
CBAM	5.0	68.91
DFC	5.1	69.09
MHSA	5.8	67.57
CADFC	5.2	69.69

Table 5
Impact of patch size.

Patch size	#Params	Top-1 ACC. (%)
(2,2)	5.1M	76.8
(3,3)	5.1M	76.5
(4,4)	5.1M	76.4

Table 6
Impact of CADFC low-rank convolution kernel size.

Kernel size	#Params (M)	Top-1 ACC.(%)
(3,3)	5.18M	76.2
(5,5)	5.18M	76.3
(7,5)	5.19M	76.6
(7,7)	5.19M	76.8

width h and w, respectively, when extracting the global information. The depthwise convolution in MV2 extracts local information from $n \times n$ regions, which is usually set to 3×3. We conducted three sets of experiments to investigate the impact of patch size on network performance, as follows:

(1) $h = w = 2 < n$, where the patch size is smaller than the receptive field of the convolution kernel.
(2) $h = w = n = 3$, where the patch size is equal to the receptive field of the convolution kernel.
(3) $h = w = 4 > n = 3$, where the patch size is larger than the receptive field of the convolution kernel.

Our experimental results, which are presented in Table 5 and are consistent with those in [39], show that when the patch size is smaller than the receptive field of the convolution kernel (as shown in Fig. 5 (b)), the network can more efficiently integrate both local and global information. Similarly, when the patch size is equal to the receptive field of the convolution kernel (as shown in Fig. 5(c), the locally extracted information by the convolution can be fully fused with the globally captured information by CADFC. However, when the patch size is larger than the receptive field of the convolution kernel (as shown in Fig. 5 (a)), the convolution cannot cover all the information in the pixels learned by patches, resulting in a decline in network performance. In actual inference, the computation complexity of CADFC is $O\left(\frac{(HW)^2}{hw} + HW \times hw\right)$, where HW is the height and width of the input feature map. By setting the patch size to 2, we can reduce the computational overhead of CADFC, and therefore, we chose a smaller patch size to improve the network's efficiency.

The impact of kernel size in CADFC attention. We investigated the impact of convolution kernel size on the performance of CADFC. As shown in Table 6, when using 1×3 and 3×1 convolution kernels to construct attention maps, the network was unable to effectively learn long-range dependencies between pixels. With the increase of kernel size, CADFC was able to better capture the long-range dependencies between pixels and achieved a significant improvement in performance.

Inspecting of CADFC MobileNet. To gain a deeper understanding of how CADFC MobileNet processes images and makes decisions, we visualized the model's response to input images and highlighted the regions relevant to the predicted class. The results of Grad-CAM clearly show the areas of interest. The intensity of the red color in the displayed regions indicates the level of attention from the model. We compare the visualization results of CADFC and DFC. As shown in Fig. 6, we can clearly observe that the Grad-CAM mask of CADFC covers the target regions better than DFC. That is, CADFC, which can profitably utilize the information in target regions, aggregate features from them. The model captures local features of the dog, such as the mouth, eyes, and ears. Additionally, the model also pays attention to the overall features of the dog. This suggests that the model considers not only the local features of objects in the image but also the global features when performing image classification. It makes classification decisions based on the extracted local and global information.

To further understand the image processing and decision-making capabilities of CADFC MobileNetNet, we visualized the attention maps generated by the CADFC and compared them with DFC. Fig. 7 illustrates the differences between CADFC and DFC. Compared to DFC, the attention map extracted by CADFC is more like the attention map generated by transformer models, as it can capture the global information of the image more comprehensively. On the other hand, DFC exhibits a noticeable criss-cross pattern, indicating less attention towards the four corners of the image.

Fig. 8 illustrates partial attention maps obtained separately by DFC and CADFC. A comparative analysis reveals attention deficiencies in the maps generated by DFC (highlighted by the red rectangle), attributed to its simplistic construction of attention maps solely from vertical and horizontal directions. Conversely, the attention maps derived from CADFC exhibit smoother transitions in the focused regions, indicating its superior compression of global image information.

The impact of horizontal and vertical. The process of constructing global attention maps in CADFC can be divided into two steps, which involve building global and local relationships between pixels in both horizontal and vertical directions. The specific impact of the two steps on the final model performance is shown in Table 7. When constructing attention maps only from the horizontal or vertical direction, it is impossible to capture the complete relationships between pixels. Especially when constructing attention maps only from the vertical direction, the performance improvement is limited because the relationships between pixels within a patch are only captured, without the

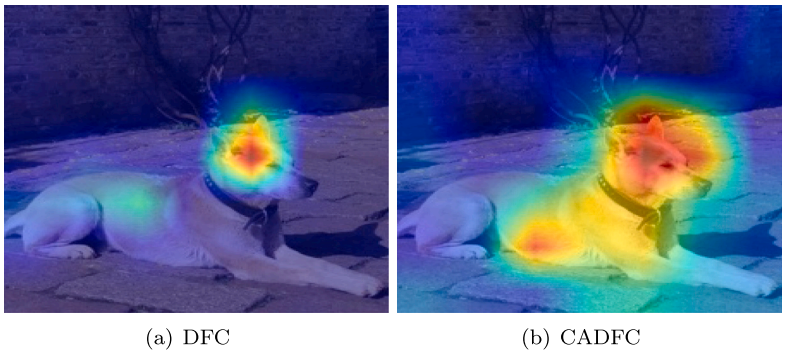


Fig. 6. (a) and (b) represent Grad-CAM of the DFC and CADFC, respectively.



Fig. 7. Visualization of attention maps.

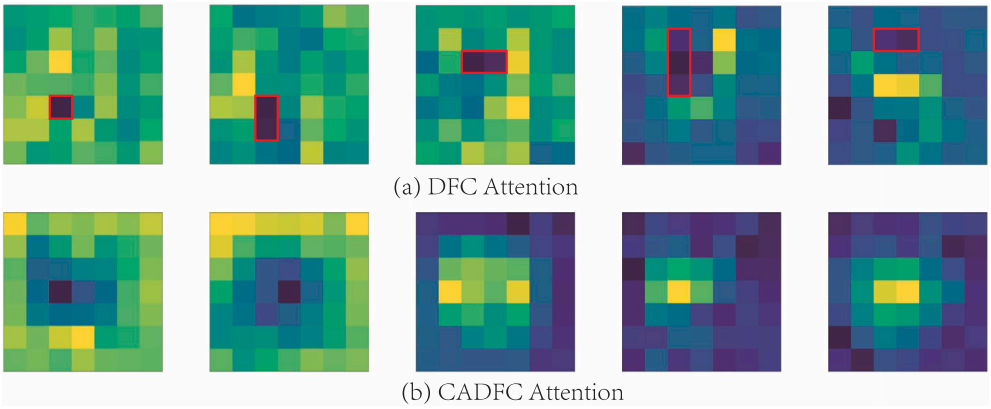


Fig. 8. The comparison of attention maps obtained by CADFC and DFC.

Table 7
Impact of attention map construction.

Model	Top-1 (%)	Inference time (ms)
With H and V	76.8	0.517
With H	76.5	0.504
With V	76.3	0.507
Without	76.1	0.289

Table 8
Impact of kernel size.

Kernel size	#Params(M)	Top-1 (%)
(3,3)	5.1	76.8
(5,5)	5.2	76.5
(7,7)	5.2	76.3
(9,9)	5.2	76.3

global information. By constructing attention maps in both horizontal and vertical directions, the resulting attention maps can capture both the global information of the image and the local information of the patch, leading to further performance improvement of the network.

The impact of depthwise kernel size. This study conducted four sets of comparative experiments to assess the influence of different sizes of depthwise convolutional kernels on the performance of the CADFC MobileNet Bottleneck. The experimental results are presented in Table 8. reveal that the inclusion of larger convolutional kernels does not contribute to an enhancement in network performance. Instead, it leads to a decline in performance as the expanded receptive field of the convolutional kernels causes the loss of local information in the feature maps.

6. Conclusion

This study introduces a small-scale architecture based on CNNs enhanced by a Context-Aware Decoupled Fully Connected (CADFC) attention mechanism. This mechanism effectively integrates both remote and local contexts, allowing the network to capture local features

and remote features. Furthermore, we have designed an innovative bottleneck that combines CADFC attention with depthwise convolution, facilitating the accumulation of focused regions alongside their surrounding features. Experimental results demonstrate that CADFC MobileNet outperforms the recent SOTA network GhostNetV2, achieving a top-1 accuracy of 76.8% with 15% fewer parameters, surpassing GhostNetV2's 75.3% accuracy by 1.5%. Additionally, it shows an average improvement of 13.85% in AP for object detection, even in resource-constrained environments. This approach underscores the potential of lightweight visual networks for effective real-time decision-making in robotics and drone applications.

CRediT authorship contribution statement

Jianxin Xue: Methodology, Formal analysis, Conceptualization. **Yaohua Hu:** Methodology. **Sicheng Hua:** Software. **Minyu Chen:** Validation. **Ling-I Wu:** Validation. **Xi Chang:** Writing – original draft. **Guoqiang Li:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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